QRM II Graded Assignment (14)

Period 1, 2026

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0 Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so **before the start of your first R tutorial on Friday September 4th.** You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is **not** permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show how you analyzed the data to get the answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

Each week consists of 1, 2, or 3 subquestions. The total amount of points you can earn per week is 20 points.

1 Week 1

1. Find the dataset "movies1.tsv" on Canvas. Describe your data set: How many observations does it have. How many variables are there? How many subjects? What consists of a subject? [**4 points**]

```
library(readr)
movies1 <- read_tsv("movies1.tsv", show_col_types = FALSE)
nrow(movies1)
```

```
## [1] 505
```

```
ncol(movies1)
```

```
## [1] 19
```

Your Answer: The data set has 505 observations and 19 variables. Each observation is one movie, so there is 505 subjects, and a subject is a single movie.

2. Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii); interval; (iv) ratio. If present, name one example of such a variable present in your data set. [**4 points**]

```
head(movies1[, c("genre", "release_month", "release_date", "budget")], 3)
```

```
## # A tibble: 3 x 4
##   genre    release_month release_date   budget
##   <chr>            <dbl> <date>          <dbl>
## 1 Thriller             5 2004-05-07   27000000
## 2 Comedy              11 2015-11-26   15000000
## 3 Comedy               1 2016-01-14   40000000
```

Your Answer:

All four data types are present in the data set:

- Nominal: genre
- Ordinal: release_month
- Interval: release_date
- Ratio: budget

3. A movie studio wants to know which types of movies give maximal profit. Perform the following steps to provide the movie studio with an analysis which corresponds to their request:

a. Create the variable profits as the revenue of a movie minus its budget. Report its mean, median, maximum, and minimum. [**2 points**]

b. Which movie has the highest profits in your data set and how much are these profits. Which movie has the lowest and how much are its profits? If multiple movies have the exact same highest or lowest profits, give only one example. [**2 points**]

c. Create a boxplot of the variable profits. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Give Q1, Q2, Q3, and Q4. What does this tell you about the nature of making money in the movies industry? [**2 points**]

d. Add a new variable to your data set the log of profits. When creating this variable, what happens to movies for which profits is zero or negative? What then happens when you calculate the mean of log of profits? [**2 points**]

e. For movies that have a profit of zero or less, replace log of profits with "NA". What is now the mean of log of profits? Create a boxplot for log of profits, again with an appropriate title, x- and y-axis labels. How does it compare to the boxplot you made under c.)? [**2 points**]

f. Create a scatterplot of with the runtime of movies on the x-axis and the average vote of movies on the y-axis. What do you conclude from the scatterplot? Are movies with a longer runtime considered worse or better by the audience, or does the audience not have a preference? Why do you think this is the case? [**2 points**]

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Step a

```
library(readr)

movies1 <- read_tsv("movies1.tsv", show_col_types = FALSE)

movies1$profits <- movies1$revenue - movies1$budget

mean(movies1$profits)
```

```
## [1] 63121475
```

```
median(movies1$profits)
```

```
## [1] 1900000
```

```
max(movies1$profits)
```

```
## [1] 2550965087
```

```r
min(movies1$profits)
```

```
## [1] -9e+07
```

Your Answer:

After creating profits as revenue minus budget, the average profit across the 505 movies is $63,121,475, but the median is only $1,900,000. Profits range from a low of -$90,000,000 (a loss) up to a high of $2,550,965,087. The big gap between the mean and median already suggests the data is skewed, since a few really successful movies are dragging the average up while most movies make much less.

Step b

```r
movies1$profit <- movies1$revenue - movies1$budget

highest_profit_movie <- movies1[which.max(movies1$profit), ]
highest_profit_movie[, c("title", "profit")]
```

```
## # A tibble: 1 x 2
##   title      profit
##   <chr>       <dbl>
## 1 Avatar 2550965087
```

```r
lowest_profit_movie <- movies1[which.min(movies1$profit), ]
lowest_profit_movie[, c("title", "profit")]
```

```
## # A tibble: 1 x 2
##   title               profit
##   <chr>                <dbl>
## 1 Mighty Joe Young -90000000
```

Your Answer:

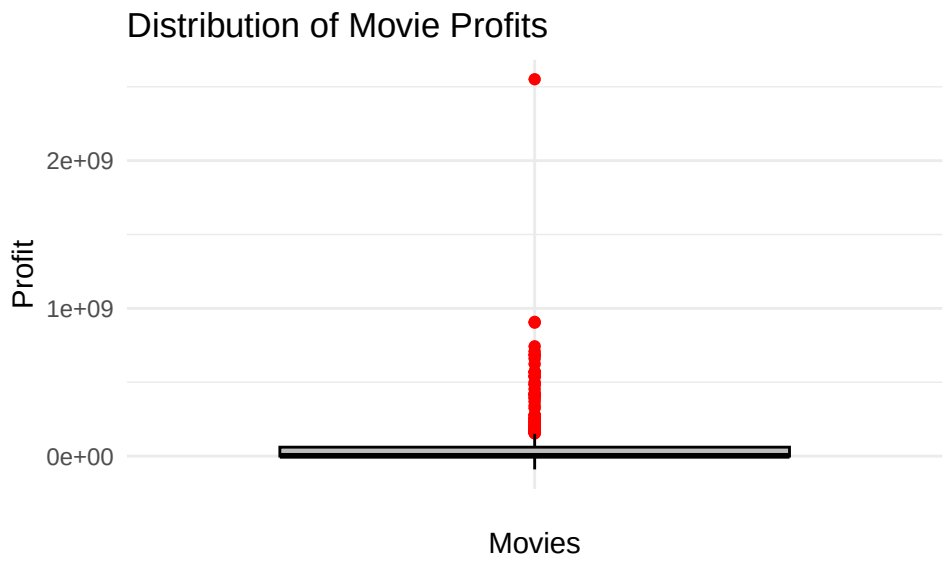
The highest profit is Avatar, which has a profit of 2,550,965,087 dollars The lowest profit is Mighty Joe Young, which has a loss of 90,000,000 dollars

Step c

```r
library(ggplot2)
movies1$profit <- movies1$revenue - movies1$budget
quantile(movies1$profit, na.rm = TRUE)
```

```
##         0%        25%        50%        75%       100%
##  -90000000          0    1900000   60514050 2550965087
```

```r
ggplot(movies1, aes(x = "", y = profit)) +
  geom_boxplot(fill = "gray", color = "black", outlier.color = "red") +
  labs(
    title = "Distribution of Movie Profits",
    x = "Movies",
    y = "Profit"
  ) +
  theme_minimal()
```



Your Answer: Q1 = \$0, Q2 (median) = \$1,900,000, Q3 = \$60,514,050 and Q4 (maximum) = \$2,550,965,087. The movie industry is very high risk high reward, whilst most movies either break even or make a small profit, or a small loss, only a low amount of movies actually make a large profit

Step d

```
movies1$logprofits <- log(movies1$profits)
```

```
## Warning in log(movies1$profits): NaNs produced
```

```
sum(movies1$profits == 0)
```

```
## [1] 113
```

```
sum(movies1$profits < 0)
```

```
## [1] 116
```

```
sum(movies1$profits <= 0)
```

```
## [1] 229
```

```
mean(movies1$logprofits)
```

```
## [1] NaN
```

Your Answer:

Taking the log of profits gives -Inf for movies with zero profit and NaN for movies with a loss, since you can't take the log of zero or a negative number. This affects 229 of the 505 movies (113 with zero profit, 116 with a loss). Because of these values, the mean of logprofits also comes out as NaN.

Step e

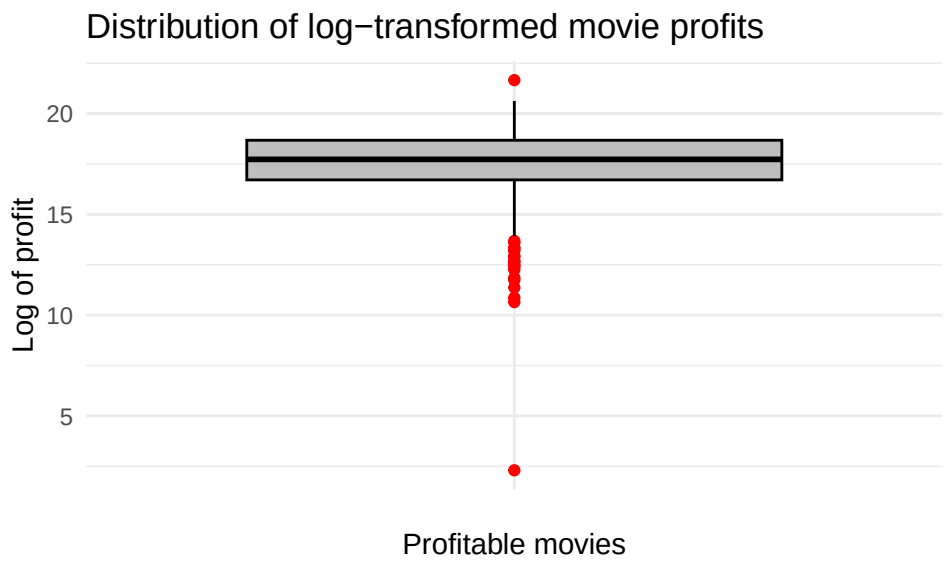
```r
movies1$profit <- movies1$revenue - movies1$budget
movies1$log_profit <- log(movies1$profit)
```

```
## Warning in log(movies1$profit): NaNs produced
```

```r
movies1$log_profit[movies1$profit <= 0] <- NA
mean_log_profit <- mean(movies1$log_profit, na.rm = TRUE)
mean_log_profit
```

```
## [1] 17.38094
```

```r
ggplot(movies1[!is.na(movies1$log_profit), ], aes(x = "", y = log_profit)) +
  geom_boxplot(fill = "gray", color = "black", outlier.color = "red") +
  labs(
    title = "Distribution of log-transformed movie profits",
    x = "Profitable movies",
    y = "Log of profit"
  ) +
  theme_minimal()
```

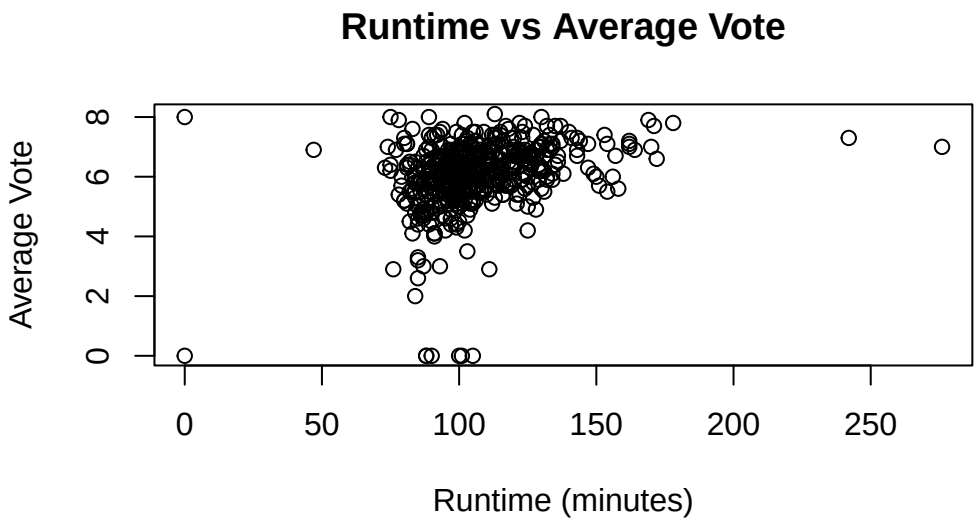


Your Answer:

After setting log profits to NA for movies with zero or negative profit, the mean of log profits is 17.38. The boxplot in C had a lot of massive outliers, whilst this one is a bit more centered around its median. Plus the boxplot in C had a lot more movies, since we deleted all the once making a loss or breaking even.

Step f

```r
plot(movies1$runtime, movies1$vote_average,
     main = "Runtime vs Average Vote",
     xlab = "Runtime (minutes)",
     ylab = "Average Vote")
```




```
cor(movies1$runtime, movies1$vote_average, use = "complete.obs")
```

```
## [1] 0.3211682
```

This gives a correlation of about 0.32 between runtime and average vote. Looking at the scatterplot, most movies cluster between about 90 and 130 minutes with average votes somewhere between 5 and 7.5, and there's a mild upward trend running through the cloud of points. Longer movies do tend to get slightly higher ratings on average, but the relationship is pretty weak, since a correlation of 0.32 means runtime only accounts for a small part of the variation in how movies get rated.

So overall, audiences seem to have a slight preference for longer movies, but it's not a strong effect. A likely reason is that longer runtimes often go along with bigger productions, more developed stories, and higher budgets, things like epics or dramas that studios invest more time and money into tend to run longer, and that extra effort may translate into somewhat better reception.

2 Week 2

1 Is your dataset movies1.tsv the full population, or is it a sample of a larger population? If the latter, how would you describe the full population? [**4 points**]

Your Answer:

It is a sample of a larger population, the full population would include every single movie made globally, and since this dataset only includes 505 movies, it is a sample.

2 a. For which actor in your data set do you observe the most movies? [**2 points**] b. What is the average revenue of the movie in which this actor plays and does the revenue lie above or below the revenue of an average movie according to your data set? [**2 points**]
c. How trustworthy do you consider your conclusion to answer 2b? Use the term "law of large numbers" in your explanation. [**2 points**]

step a

```r
library (dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
movies1 %>%
  filter(!is.na(first_actor)) %>%
  count(first_actor, sort = TRUE) %>%
  head(1)
```

```
## # A tibble: 1 x 2
##   first_actor      n
##   <chr>        <int>
## 1 Bruce Willis     7
```

Your Answer: The actor with the most movies (in this sample) is Bruce Willis who has 7 movies (in this sample).

step b

```r
movies_bruce <- subset(movies1, first_actor == "Bruce Willis")
Average_revenue_Bruce <- mean(movies_bruce$revenue, na.rm = TRUE)

Average_Revenue <- mean(movies1$revenue, na.rm = TRUE)
Average_revenue_Bruce
```

```
## [1] 116280090
```

```r
Average_Revenue
```

```
## [1] 94815482
```

Your Answer:

The average revenue for a Bruce Willis film is 116,280,089.57 U.S. Dollars The average revenue for all the movies is 94,815,481.64 U.S. Dollars

This means that the average Bruce Willis movie generates more revenue than the average movie from this sample.

step c

Your Answer:

It is not very trustworthy since this sample is only a small percentage of all movies made. According to the law of large numbers, the more movies you add to the sample, the closer you would get to the 'true mean'. And since we are only analyzing 505 movies and 7 Bruce Willis movies, the possibility is very large that the answer is distorted. (e.g. there could be one massive outlier which manipulates the outcome)

3 For this question, you will assume that your data set is the full population.

a. Recode profits such that it is expressed in millions. What is the variance of the variable profits (in millions) in your data set? [**2 points**]
b. Create a new data set, called movies_sample. Make sure that it is a random sample of your data set of 25 movies. What is the variance of profits in this random sample? How does it compare to the variance of profits in 2a? [**2 points**]
c. In a for loop, create 100 different samples of 25 movies, as in b, and estimate the variance within each sample. Save the variance of each sample in a vector called sample_vars. So the first position of the vector would have the variance of the first sample, the second position the variance of the second sample, etc. Print the start of this vector. [**2 points**]
d. Summarize and make a histogram of sample_vars. What is the mean, standard deviation and shape of its distribution? [**2 points**]
e. In your opinion, is a sample of 25 movies sufficient to get a reliable estimate of the population variance of profits, using the sample variance? Explain? [**2 points**]

step a

```
movies1$profits <- movies1$revenue - movies1$budget
movies1$profits_millions <- movies1$profits / 1e6

n <- nrow(movies1)
pop_var <- var(movies1$profits_millions, na.rm = TRUE) * (n - 1)/n
pop_var
```

```
## [1] 30402.8
```

Your Answer: After recoding profits into millions and treating the dataset as the full population, the population variance of profits is 30,402.8 (million dollars squared). This gives a standard deviation of about $174 million, which is large compared to the mean profit of roughly $63 million. This makes sense given the skewed nature of the data, since a small number of blockbuster movies like Avatar generate profits far above the typical movie, which pulls the variance up a lot.

step b

```
set.seed(123)
movies_sample<- movies1[sample(nrow(movies1), 25),]

sample_var <- var(movies_sample$profits_millions, na.rm = TRUE)
sample_var
```

```
## [1] 5156.149
```

Your Answer:

The variance of profits in the random sample of 25 movies is 5156.149 (million dollars squared), which is much lower than the population variance of 30,402.8 found in 2a. This difference is likely because the profits distribution is highly skewed, with a small number of blockbuster movies driving up the variance in the full

population. A sample of just 25 movies has a good chance of missing these extreme outliers entirely, which would make the sample variance look much smaller than the true population variance. This shows that with such a skewed variable, small samples can give quite unreliable estimates of the population variance, just by chance.

step c

```
set.seed(2026)
sample_vars <- numeric(100)

for (i in 1:100) {
sample_i <- movies1[sample(nrow(movies1), 25), ]
sample_vars[i] <- var(sample_i$profits_millions, na.rm=TRUE)
}
head(sample_vars)
```

```
## [1] 20037.60 20651.72 51398.94 36983.01 18170.74 14105.60
```

Your Answer:

The first six values are 20037.60, 20651.72, 51398.94, 36983.01, 18170.74, and 14105.60. These values already show a lot of variation from sample to sample, which reflects how sensitive the variance estimate is to which movies happen to end up in a small sample of 25, especially given how skewed the underlying profits distribution is.

step d

```
library(ggplot2)
mean_var <- mean(sample_vars)
sd_var <- sd(sample_vars)

mean_var
```

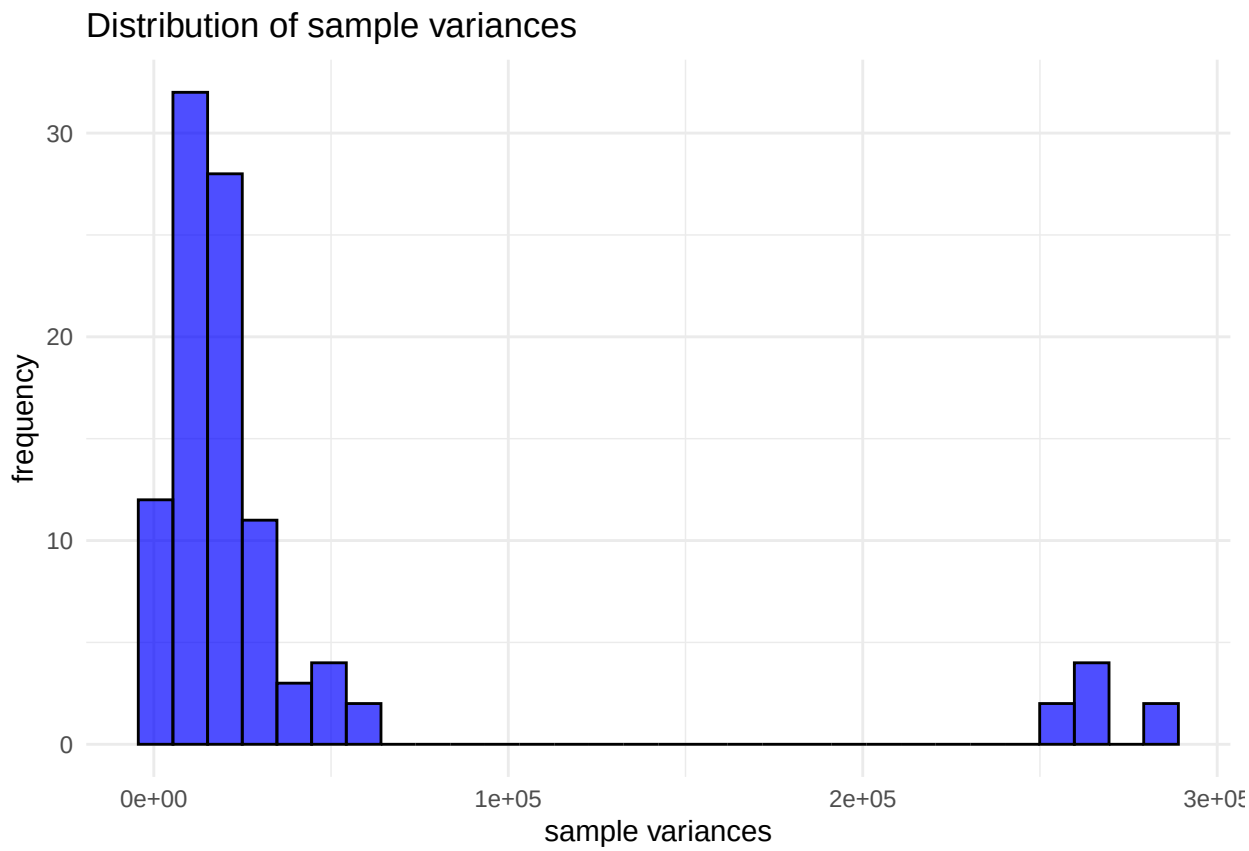
```
## [1] 38329.65
```

```
sd_var
```

```
## [1] 68929.61
```

```
df_vars <- data.frame(sample_vars = sample_vars)

ggplot(df_vars, aes(x = sample_vars)) +
  geom_histogram(bins = 30, fill = "blue", color = "black", alpha = 0.7) +
  labs(
    title = "Distribution of sample variances",
    x = "sample variances",
    y = "frequency"
  ) +
  theme_minimal()
```



Your Answer:

The mean of sample_vars is about 38,330 and its standard deviation is about 68,930 (both in millions of dollars squared). The mean is in the same range as the population variance of 30,402.8, but not equal to it. The sample variance is an unbiased estimator, so with more repetitions the average would move closer to the population value. However, the standard deviation is more than twice the mean, which shows how much the estimates vary between samples. The histogram is strongly right skewed. Most samples have a variance well below the population variance (the median is only about 16,880), and a few samples that happen to include a blockbuster such as Avatar have extremely large variances (up to about 284,000).

step e

Your Answer:

No, a sample size of 25 is not sufficient. This is because a single mega hit amongst those 25 can extremely influence the variance of the sample

3 Week 3

For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.

1

a. Create a new data set that only includes movies that are of the genre "Thriller". For these thriller movies, give a 99 percent confidence interval for the variable *runtime*. Interpret the result. **[2 points]**

b. Now, assume that the variance of *runtime* amongst thriller movies in your data is exactly the same as the variance of *runtime* in the population. Under this assumption, give a 99 percent confidence interval for the variable *runtime* among thriller movies. Interpret the result. Is you confidence interval wider or less wide than the one you found under question 1a? Why is that the case? **[2 points]**

step a

```
thrillers <- subset(movies1, genre == "Thriller")
xbar  <- mean(thrillers$runtime, na.rm = TRUE)
s     <- sd(thrillers$runtime, na.rm = TRUE)
n     <- sum(!is.na(thrillers$runtime))
se    <- s / sqrt(n)
tcrit <- qt(0.995, df = n - 1)
c(xbar - tcrit * se, xbar + tcrit * se)
```

```
## [1]  99.94343 111.01042
```

Your Answer:

We're 99% confident the true mean runtime of all thriller movies lies between 99.94 and 111.01 minutes. The interval is centered on the sample mean of 105.48 minutes, based on the 65 thrillers in our data. Because the population variance is unknown, we estimate it with the sample variance and use the t distribution with 64 degrees of freedom. We do not need runtime itself to be normally distributed, since $n = 65 \geq 30$, the central limit theorem tells us that the sample mean is approximately normal, so this interval is valid.

step b

```
zcrit <- qnorm(0.995)
c(xbar - zcrit * se, xbar + zcrit * se)
```

```
## [1] 100.1081 110.8457
```

Your Answer:

The 99% confidence interval is [100.11, 110.85]. Assuming the population variance is known, we are 99% confident that the true mean runtime of all thriller movies lies between 100.11 and 110.85 minutes. This interval is narrower than the one in part a. Because the variance is now known, we no longer have the extra uncertainty from estimating it with the sample, so we use the z (normal) distribution instead of the t distribution. Its critical value is smaller (2.576 instead of 2.655), which gives a narrower interval. The difference is small because with n = 65 the t distribution is already very close to the normal distribution.

2

a. Using an appropriate five-step procedure, set up a test for the null hypothesis that the variance of runtime equals 500. Clearly state your null hypothesis, alternative hypothesis your test statistic, your critical value, and your conclusion. **[2 points]**
b. For the validity of your test in 2a, what assumption about the distribution of runtime needs to hold? Make an appropriate plot to test this assumption. What do you conclude? **[2 points]**

step a

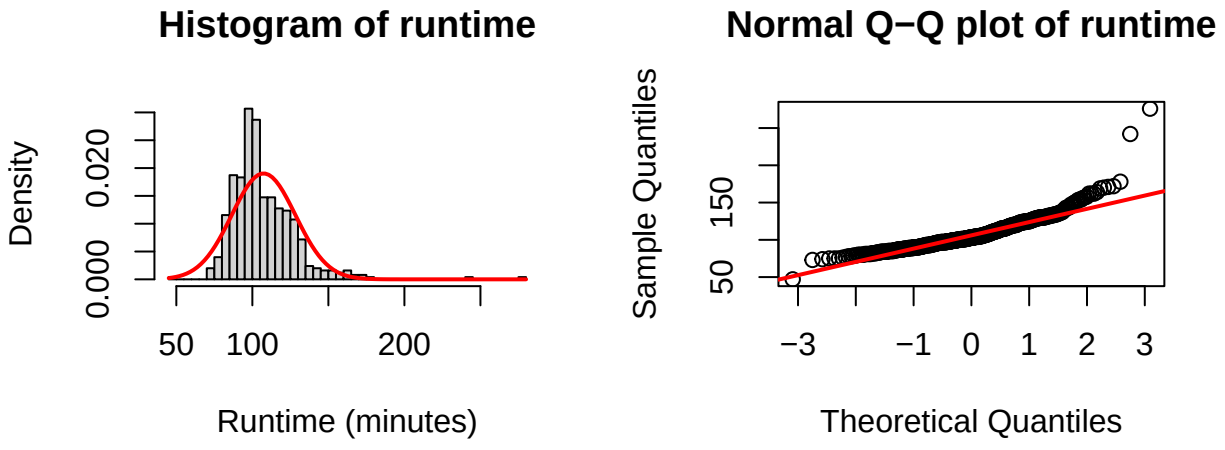
```r
x <- movies1$runtime[!is.na(movies1$runtime) & movies1$runtime > 0]
n  <- length(x)
s2 <- var(x)
chi2 <- (n - 1) * s2 / 500
crit <- qchisq(c(0.025, 0.975), df = n - 1)
pval <- 2 * min(pchisq(chi2, n - 1), 1 - pchisq(chi2, n - 1))
round(c(n = n, s2 = s2, chi2 = chi2, crit_low = crit[1], crit_high = crit[2], p = pval),
↪  3)
```

```
##         n        s2      chi2  crit_low crit_high         p
##   502.000   439.178   440.057   440.874   564.914     0.047
```

1. **Hypotheses:** $H_0 : \sigma^2 = 500$ versus $H_1 : \sigma^2 \neq 500$ (two-sided), with significance level $\alpha = 0.05$.
2. **Decision rule:** we use the sample variance s^2. Because the test is two sided, we reject H_0 if s^2 is much smaller or much larger than 500, i.e. if χ^2 falls below the lower or above the upper critical value.
3. **Distribution:** under H_0, $\chi^2 = (n-1)s^2/\sigma_0^2$ follows a χ^2 distribution with $n-1$ degrees of freedom. This requires runtime to be normally distributed in the population. We removed one movie with a missing runtime and two with a runtime of 0, which is not a real value, leaving $n = 502$ movies and $n - 1 = 501$ degrees of freedom.
4. **Computation:** $s^2 = 439.18$, so $\chi^2 = 501 \cdot 439.18/500 = 440.06$. The critical values are 440.87 (lower) and 564.91 (upper), and the p value is 0.047.
5. **Decision:** $\chi^2 = 440.06$ is below the lower critical value and $\text{p} = 0.047 < 0.05$, so we reject H_0. At the 5% level there is evidence that the population variance of runtime differs from 500. The rejection is borderline, though, with the two zero runtimes included we would not reject, so this result should be read with care.

step b

```r
par(mfrow = c(1, 2))
hist(x, breaks = 40, freq = FALSE, main = "Histogram of runtime",
     xlab = "Runtime (minutes)")
mu   <- mean(x)
sdev <- sd(x)
curve(dnorm(x, mean = mu, sd = sdev), add = TRUE, col = "red", lwd = 2)
qqnorm(x, main = "Normal Q-Q plot of runtime")
qqline(x, col = "red", lwd = 2)
```



```r
par(mfrow = c(1, 1))
```

Your Answer:

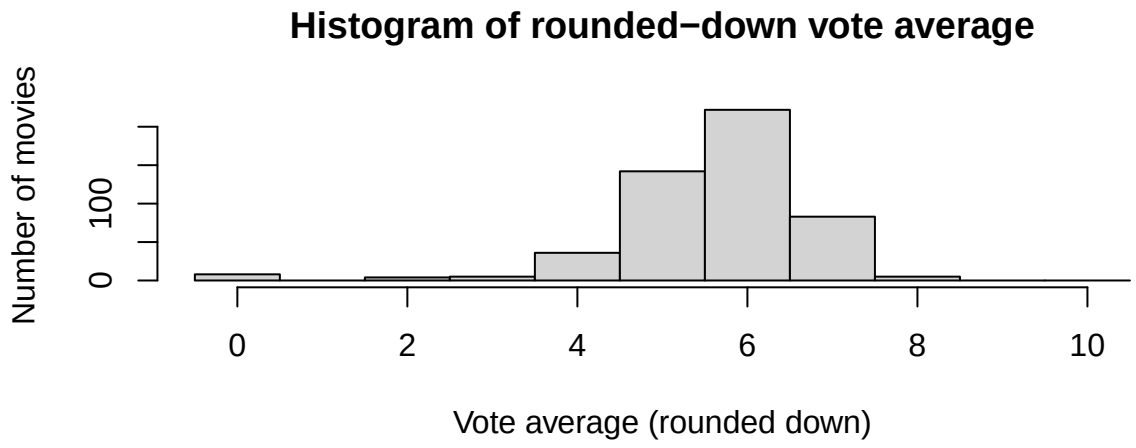
The χ^2 test for a variance is only valid if runtime is normally distributed in the population, and unlike tests on the mean it stays sensitive to this even in large samples. The histogram shows that runtime is not normal: it is right-skewed, with a long upper tail (up to 276 minutes). In the Q-Q plot the points bend away from the line in both tails, most clearly in the upper tail. The normality assumption is therefore violated. The conclusion in 2a is that H_0 was rejected because $p = 0.047 < 0.05$, but this p value comes from a test whose assumption doesn't hold, so there is no strong proof that the variance is different from 500.

3. There is an argument going on in the movie studio. *Bob* claims that they should make higher-quality movies, as this will bring in more profits. *Chantal* disagrees. She tells Bob that mediocre movies bring in the most profits. You are asked to advise on who is right.

a. Create a new variable called vote_average_rounded. Make sure this variable is the same as vote_average, but without any decimals (i.e., a 6.3 becomes a 6, a 8.7 an 8, etc.). Display a histogram of vote_average_rounded. [**2 points**]
b. Create a scatter plot with vote_average_rounded on the x axis and the mean of profits within each category of vote_average_rounded on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. At which rating of movies are profits the highest? [**3 points**]
c. Recreate the scatter plot with year on the x axis and mean_profits on the y-axis, but now add bars around each point, indicating the 95% confidence interval. [**3 points**]
d. Write an advice to settle the argument between Bob and Chantal. [**4 points**]

step a

```r
movies1$vote_average_rounded <- floor(movies1$vote_average)
hist(movies1$vote_average_rounded, breaks = seq(-0.5, 10.5, by = 1),
     main = "Histogram of rounded-down vote average",
     xlab = "Vote average (rounded down)", ylab = "Number of movies")
```



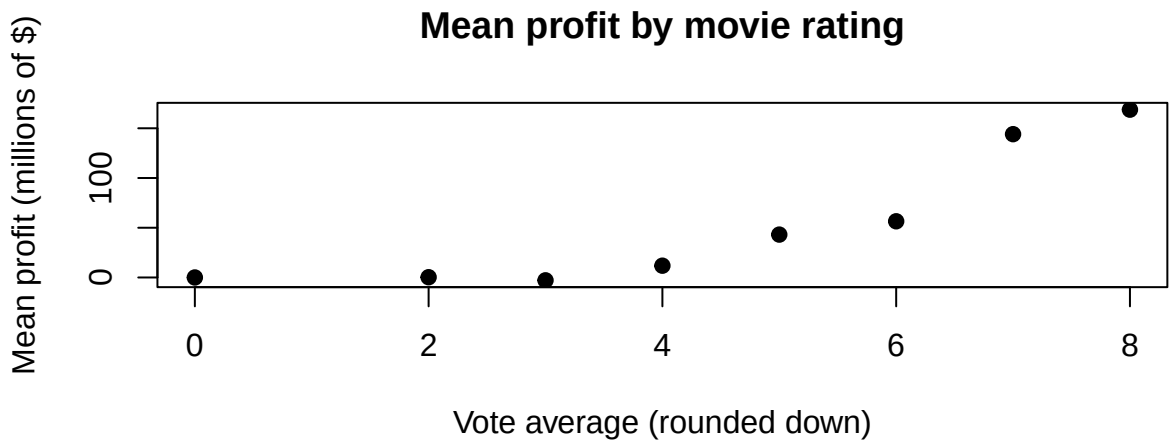
Your Answer:

We used floor() instead of round(), so decimals are simply cut off. For example, a 6.7 becomes 6, not 7. Looking at the results, most movies land in the middle, with 222 rated 6 and 142 rated 5. After that, 83

movies at 7 and 36 at 4. Top scores are rare, only 5 movies reach an 8, and not a single one gets a 9 or 10. The 8 movies with a rating of 0 all have zero votes, so that 0 doesn't mean they're terrible. It just means nobody rated them.

step b

```
mean_profits <- aggregate(profits ~ vote_average_rounded, data = movies1, FUN = mean)
plot(mean_profits$vote_average_rounded, mean_profits$profits / 1e6, pch = 19,
     main = "Mean profit by movie rating",
     xlab = "Vote average (rounded down)", ylab = "Mean profit (millions of $)")
```

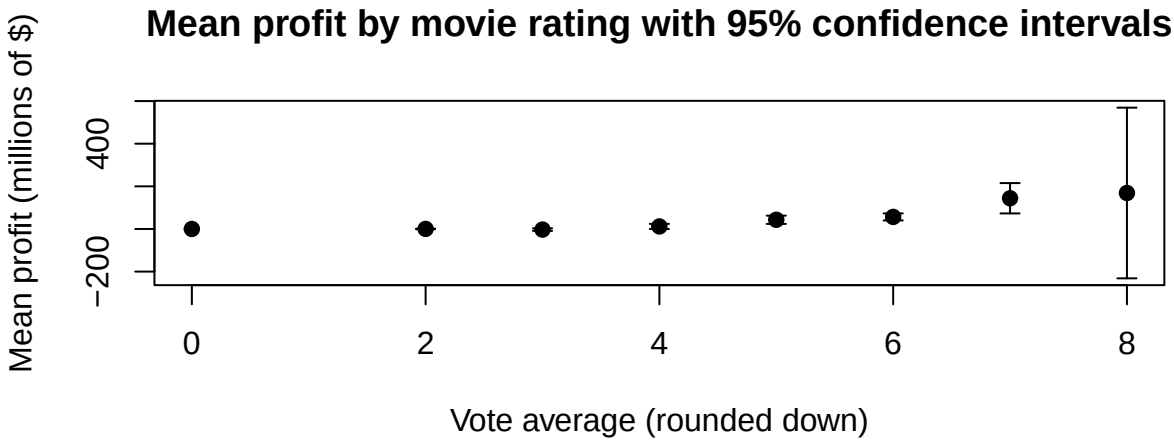


Your Answer:

Mean profits are highest for movies rated 8, at about $168.7 million, followed by movies rated 7 ($144.1 million). Profits rise steadily from rating 4 onwards, movies rated 4 make about $11.8 million, and movies rated 5 and 6 make on average $43.1 and $56.5 million. One thing worth mentioning is that movies with a rating of 3 lose money, about $2.9 million on average.

step c

```
ci <- do.call(data.frame, aggregate(profits ~ vote_average_rounded, data = movies1,
        FUN = function(v) c(mean = mean(v), sd = sd(v), n = length(v))))
names(ci) <- c("rating", "mean", "sd", "n")
ci$lower <- (ci$mean - qt(0.975, ci$n - 1) * ci$sd / sqrt(ci$n)) / 1e6
ci$upper <- (ci$mean + qt(0.975, ci$n - 1) * ci$sd / sqrt(ci$n)) / 1e6
ci$mean  <- ci$mean / 1e6
plot(ci$rating, ci$mean, pch = 19, ylim = range(c(ci$lower, ci$upper)),
     main = "Mean profit by movie rating with 95% confidence intervals",
     xlab = "Vote average (rounded down)", ylab = "Mean profit (millions of $)")
arrows(ci$rating, ci$lower, ci$rating, ci$upper, angle = 90, code = 3, length = 0.05)
```




```
round(ci[, c("rating", "n", "mean", "lower", "upper")], 1)
```

```
##   rating   n  mean  lower upper
## 1      0   8   0.0    0.0   0.0
## 2      2   4   0.2   -1.1   1.6
## 3      3   5  -2.9   -8.7   3.0
## 4      4  36  11.8   -0.3  23.9
## 5      5 142  43.1   23.7  62.6
## 6      6 222  56.5   40.1  72.9
## 7      7  83 144.1   72.9 215.3
## 8      8   5 168.7 -231.6 569.1
```

Your Answer:

We read "year" in the question as vote_average_rounded, since the question asks us to recreate the plot from 3b. For ratings 5 and 6 the intervals are fairly narrow ($23.7–62.6 million and $40.1–72.9 million), which makes sense because these groups contain the most movies. Rating 7 is the most interesting, its interval runs from $72.87 to $215.3 million, which sits entirely above the interval for rating 5 and only barely overlaps with the one for rating 6 (which ends at $72.89 million). Finally, the interval for rating 8 is huge (–$231.6 to $569.1 million) because it's based on just 5 movies, so we can't put much trust in its high mean. The interval even dips into negative values, so we can't be sure these movies are profitable on average at all. More generally, these t-intervals rely on the central limit theorem, which needs roughly 30 or more observations per group. Ratings 0, 2, 3 and 8 contain only 8, 4, 5 and 5 movies, and profits are very right skewed, so the intervals for these groups are not reliable.

step d

Your Answer:

The data support Bob rather than Chantal. Mediocre movies (ratings 5–6) earn on average $43–56 million, while good movies rated 7 earn about $144 million. The 95% confidence interval for rating 7 lies entirely above the one for rating 5 and only barely overlaps with the one for rating 6 (which ends at $72.89 million). Movies rated 8 have the highest mean profit, but with only 5 such movies this estimate is too uncertain to rely on. There is no evidence that mediocre movies are the most profitable.

That said, Chantal has a point if she means total profit, because most movies are mediocre, ratings 5–6 together bring in about $18.7 billion, compared with about $12.0 billion for rating 7 (mean × number of movies from the table in 3c). But this only reflects how many mediocre movies are made. Per movie, a

rating 7 film earns more than twice as much, and that is what matters when the studio decides what kind of movie to make next.

We would still add two caveats. First, this is a correlation, not proof that quality causes profit, better rated movies may also have bigger budgets, famous casts, franchises or more marketing, and popular movies may attract higher ratings rather than the other way around. Second, many movies in the data have a budget or revenue of 0, which is probably missing data and distorts the profit figures. Our advice is that the studio should aim for well received (7+) movies, and confirm the effect of quality with an analysis that controls for budget and genre.

4 Week 4

1. There is another argument going on in the movie studio. *Bob* claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. *Chantal* disagrees. She tells Bob that "Every dollar we spend on movie production is more than offset by the increase in movie profits''.

   a. Set up a regression model to test Chantal's claim, and estimate it. That is, estimate:

   $$\text{Profits}_i = \beta_0 + \beta_1 \text{Budget}_i + \varepsilon_i.$$

   Print a summary of your estimated model. **[2 points]**
   b. What is the estimated value of β_1 and how do you interpet it? **[2 points]**
   c. Test for the null hypothesis that $\beta_1 \geq 0$. Report the p-value and state your conclusion. **[2 points]**
   d. Next, estimate the model

   $$\text{Log Profits}_i = \beta_0 + \beta_1 \text{Log Budget}_i + \varepsilon_i.$$

   When creating the variables Log Profits and Log Budget, make sure that movies with a Revenue or Budget of zero are assigned the value "NA". Print a summary of your estimated model **[2 points]**
   e. What is the estimated value of β_1 and how do you interpet it? **[2 points]**
   f. Which model has better fit? The level-level model or the log-log model? Explain. **[2 points]**
   g. Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? **[2 points]**

step a

```
model_chantal <- lm(profit ~ budget, data = movies1)

summary(model_chantal)
```

```
##
## Call:
## lm(formula = profit ~ budget, data = movies1)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -420678243  -37658344   14833690   17600279 1964947903
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.760e+07  7.165e+06  -2.456   0.0144 *
## budget       2.547e+00  1.302e-01  19.566   <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 131700000 on 503 degrees of freedom
## Multiple R-squared:  0.4322, Adjusted R-squared:  0.431
## F-statistic: 382.8 on 1 and 503 DF,  p-value: < 2.2e-16
```

Your Answer: We estimate $\text{Profits}_i = \beta_0 + \beta_1 \text{Budget}_i + \varepsilon_i$. Because profit is revenue minus budget, the cost of each dollar spent is already subtracted. So Chantal's claim that every dollar spent is more than offset means that profits still rise when the budget rises, i.e. $\beta_1 > 0$. This gives $H_0 : \beta_1 \leq 0$ and $H_1 : \beta_1 > 0$.

step b

```
coef(model_chantal)["budget"]
```

```
##   budget
## 2.546909
```

Your Answer:

The estimated value of β_1 is 2.547. This means that, on average, every extra dollar spent on a movie's budget is associated with an increase in profits of about \$2.55. Since profit is revenue minus budget, this corresponds to about \$3.55 of extra revenue per dollar of budget. This is an association, not necessarily a causal effect: bigger-budget movies may also differ in other ways, such as franchises, famous casts or marketing.

step c

```
t_val <- summary(model_chantal)$coefficients["budget", "t value"]
qt(0.05, df = 503)
```

```
## [1] -1.647889
```

```
pt(t_val, df = 503)
```

```
## [1] 1
```

```
pt(t_val, df = 503, lower.tail = FALSE)
```

```
## [1] 4.136684e-64
```

Your Answer:

H0: beta1 >= 0 versus H1: beta1 < 0 (one-sided, alpha = 0.05). The test statistic is t = 2.547 / 0.130 = 19.57 with 503 degrees of freedom. The critical value is -1.648. Since 19.57 > -1.648 and the one-sided p-value is approximately 1, we do not reject H0. Testing Chantal's claim instead (H0: beta1 <= 0), the one-sided p-value is practically 0, so beta1 is significantly positive.

step d

```r
movies1$log_budget <- log(movies1$budget)
movies1$log_budget[movies1$budget == 0 | movies1$revenue == 0] <- NA

movies1$log_profit <- log(movies1$profit)
movies1$log_profit[movies1$budget == 0 | movies1$revenue == 0 | movies1$profit <= 0] <-
↪  NA

model_log <- lm(log_profit ~ log_budget, data = movies1)


summary(model_log)
```

```
##
## Call:
## lm(formula = log_profit ~ log_budget, data = movies1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -7.3205 -0.6734  0.2365  0.8935  3.6926
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.12020    1.06215   5.762 2.41e-08 ***
## log_budget   0.67348    0.06171  10.914  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.429 on 253 degrees of freedom
##   (250 observations deleted due to missingness)
## Multiple R-squared:  0.3201, Adjusted R-squared:  0.3174
## F-statistic: 119.1 on 1 and 253 DF,  p-value: < 2.2e-16
```

Your Answer:

Log budget and log profits are set to NA for movies with a budget or revenue of zero (181 movies), and also for the 69 movies with zero or negative profit, because the log of zero or a negative number does not exist. The model is estimated on the remaining 255 movies: $\widehat{\log(\text{Profits})} = 6.120 + 0.673 \log(\text{Budget})$.

step e

```r
coef(model_log)["log_budget"]
```

```
## log_budget
##   0.673483
```

Your Answer:

The estimated value of β_1 is 0.673. In a log-log model the coefficient is an elasticity: a 1% increase in budget is associated with an increase in profits of about 0.67%, on average. Because $0.673 < 1$, profits grow less than proportionally with the budget. Note that this model only uses the 255 movies with a positive profit, so it says nothing about the movies that made a loss.

step f

```r
summary(model_chantal)$r.squared
```

```
## [1] 0.4321711
```

```r
summary(model_log)$r.squared
```

```
## [1] 0.3201098
```

Your Answer:

The level-level model has a higher R-squared (0.432 versus 0.320), so based on R-squared it fits better. However, the comparison is not perfect, because the models have different dependent variables and samples (505 versus 255 movies), and the level-level fit is strongly influenced by outliers such as Avatar.

step g

Your Answer:

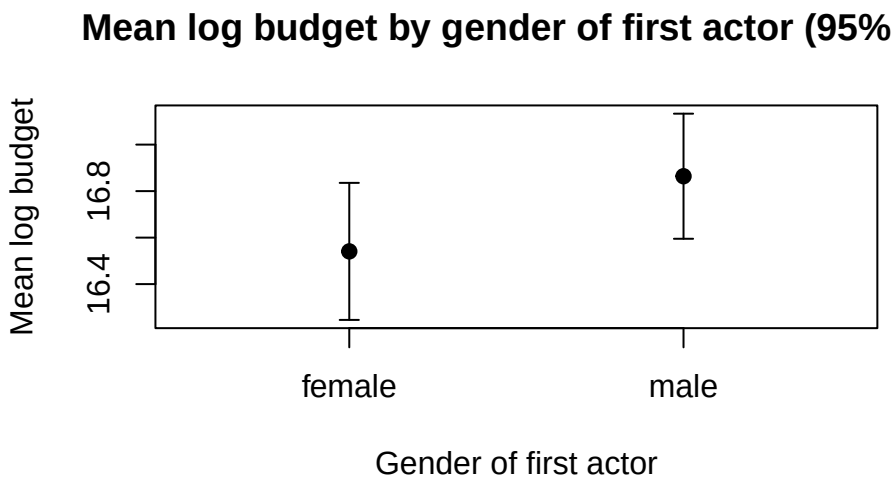
The data support Chantal: every extra dollar of budget is associated with about 2.55 dollars of extra profit, and this effect is significantly positive. The log-log model agrees (a 1% higher budget gives about 0.67% higher profits). However, this is not necessarily causal, since b1 may suffer from omitted variable bias (for example franchises, famous casts or marketing). We advise the studio not to cut budgets, but to keep investing in productions with a strong commercial case.

2

a. Make a plot with a 95% confidence interval with the mean log of budget on the y-axis, and whether the first actor of the movie is male or female on the x-axis. What do you conclude? **[2 points]**
b. Estimate the following simple OLS model: $log(budget)_i = \beta_0 + \beta_1(FirstActorMale)_i + \varepsilon_i$. Is the estimated coefficient for β_1 significantly different from zero? How do you interpret its estimate, and how does this relate to your conclusion in 2a? **[2 points]**
c. Now, have a close look at your data frame. Can you find any instances of male first actors who are wrongly labeled as being female, or vice versa? What would such mislabelling mean for the coefficient you estimated under 2b? **[2 points]**

step a

```r
movies1$log_budget2 <- ifelse(movies1$budget > 0, log(movies1$budget), NA)
gen <- subset(movies1, !is.na(first_actor_gender) & !is.na(log_budget2))
ci2 <- do.call(data.frame, aggregate(log_budget2 ~ first_actor_gender, data = gen,
         FUN = function(v) c(mean = mean(v), sd = sd(v), n = length(v))))
names(ci2) <- c("gender", "mean", "sd", "n")
ci2$lower <- ci2$mean - qt(0.975, ci2$n - 1) * ci2$sd / sqrt(ci2$n)
ci2$upper <- ci2$mean + qt(0.975, ci2$n - 1) * ci2$sd / sqrt(ci2$n)
plot(1:2, ci2$mean, pch = 19, xlim = c(0.5, 2.5), ylim = range(c(ci2$lower, ci2$upper)),
     xaxt = "n", main = "Mean log budget by gender of first actor (95% CI)",
     xlab = "Gender of first actor", ylab = "Mean log budget")
axis(1, at = 1:2, labels = ci2$gender)
arrows(1:2, ci2$lower, 1:2, ci2$upper, angle = 90, code = 3, length = 0.05)
```




```
ci2
```

```
##   gender     mean       sd   n    lower    upper
## 1 female 16.54073 1.414933  91 16.24606 16.83541
## 2   male 16.86398 2.219813 264 16.59497 17.13299
```

Your Answer:

We use only movies with a positive budget and a known gender of the first actor (91 female, 264 male). The mean log budget is 16.54 for movies with a female first actor (95% CI: 16.25 to 16.84) and 16.86 for movies with a male first actor (95% CI: 16.59 to 17.13). Movies with a male lead have a slightly higher average budget, but the two confidence intervals overlap, so there is no clear evidence of a difference in budgets between male and female first actors.

step b

```
gen$male <- ifelse(gen$first_actor_gender == "male", 1, 0)
model_gender <- lm(log_budget2 ~ male, data = gen)
summary(model_gender)
```

```
##
## Call:
## lm(formula = log_budget2 ~ male, data = gen)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.1708  -0.4835   0.3527   1.0459   2.6171
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  16.5407     0.2144   77.16   <2e-16 ***
## male          0.3232     0.2486    1.30    0.194
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 2.045 on 353 degrees of freedom
## Multiple R-squared:  0.004767,   Adjusted R-squared:  0.001948
## F-statistic: 1.691 on 1 and 353 DF,  p-value: 0.1943
```

Your Answer:

The estimated model is log(budget) = 16.54 + 0.323 * FirstActorMale. The coefficient b1 = 0.323 means that movies with a male first actor have on average a log budget that is 0.323 higher, which is roughly 32% higher budgets than movies with a female first actor. However, b1 is not significantly different from zero: t = 1.30 and p = 0.194 > 0.05, so we do not reject H0: beta1 = 0. This matches our conclusion in 2a: the intercept (16.54) is the mean log budget for female first actors, b1 is the difference between the two group means (16.86 - 16.54), and just as the confidence intervals in 2a overlapped, the regression finds no significant difference. In addition, some budgets are clearly wrong (for example 2 and 10 dollars), which are outliers that pull down the mean log budget of male-led movies and also bias the estimate.

step c

```
check <- c("Taylor Kitsch", "Jackie Chan", "Jean Reno", "Cuba Gooding",
           "Blake Lively", "Cameron Diaz", "Dakota Fanning", "Carey Mulligan")
as.data.frame(subset(movies1, first_actor %in% check)[, c("title", "first_actor",
↪  "first_actor_gender")])
```

```
##                             title    first_actor first_actor_gender
## 1                       Battleship  Taylor Kitsch             female
## 2               The Age of Adaline   Blake Lively               male
## 3                        Boat Trip   Cuba Gooding             female
## 4                     The Visitors      Jean Reno             female
## 5                      Now Is Good Dakota Fanning               male
## 6                  Never Let Me Go Carey Mulligan               male
## 7               The Sweetest Thing   Cameron Diaz               male
## 8                          Savages   Blake Lively               male
## 9                             1911    Jackie Chan             female
## 10                Shanghai Knights    Jackie Chan             female
## 11 Charlie's Angels: Full Throttle   Cameron Diaz               male
```

Your Answer:

Yes, for example Taylor Kitsch, Jackie Chan and Jean Reno are labelled as female, while Blake Lively, Cameron Diaz and Dakota Fanning are labelled as male. This means FirstActorMale is measured with error. As shown in the lecture, measurement error in X biases b1 towards zero, so the true difference in budgets may be larger than our estimate of 0.323.

5 Week 5

a. Create a plot of the mean profits by month of release. Do you see any indication that month of release matters to the profits of the movie? **[2 points]**

b. Estimate an OLS model which has as dependent variable the log of profits of a movie, and as independent variable the log of budget, a dummy for whether the movie was released in english or not, and a linear term for the month of release. Show a summary of the resulting model and interpret each coefficient. **[4 points]**

c. Test for the hypothesis that the coefficient that belongs to month of release is zero. **[2 points]**

d. Based on your plot in a.) do you consider the choice that month of release enters the model linearly under b.) reasonable? Estimate a specification that allows for a more flexible curve. In this new specification, test for the null hypothesis that month of release does not impact profits. This might require testing multiple terms at once. [**4 points**]

e. One executive at the studio wants to time the release of the movie to a specific month of the year such that they can maximize revenue. Based on your model under d.), What would you advise the movie studio regarding the timing of the release of the movie? [**2 points**]

The movie studio that you work at is releasing a new movie in 2026. It will be an English-spoken Thriller movie with a budget of 40,000,0000.

f. Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 99% prediction interval. [**6 points**]

step a

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

step b

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

step c

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

step d

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

step e

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

step f

```
#WRITE YOUR CODE HERE
```